

Honors Calculus 1610

Lecture 7: Limit Laws, Continuity, and the Intermediate Value Theorem

October 5, 2026

Indefinite integrals

The expression

$$\int_a^x f(t) dt,$$

considered as a function of x , is called an **indefinite integral** of f . (We assume throughout that the integrals exist.)

Remark

If $F(x) = \int_a^x f(t) dt$ and $G(x) = \int_b^x f(t) dt$ are indefinite integrals, then $F(x) = G(x) + C$.

Because

$$\underbrace{\int_a^x f(t) dt}_{F(x)} = \underbrace{\int_a^b f(t) dt}_{C = \text{some real number}} + \underbrace{\int_b^x f(t) dt}_{G(x)}.$$

The first term does not depend on x at all — that is the whole content.

Useful fact

If $f: [a, b] \rightarrow \mathbb{R}$ is integrable, then f is **bounded**.

That is: f has both an upper and a lower bound, so there are $U, L \in \mathbb{R}$ with

$$L \leq f(x) \leq U \quad \text{for all } x \in [a, b].$$

Proof

Say $I = \int_a^b f(x) dx$. If $Q > I$, then since

$$I = \inf \left\{ \int_a^b t(x) dx \mid t(x) \geq f(x), t \text{ a step function} \right\},$$

Q is not a lower bound, so there is a step function $t \geq f$ with $\int_a^b t(x) dx \leq Q$.

Say t is given by intervals (x_{i-1}, x_i) with $t(x) = c_i$ there, and $t(x_i) = d_i$ at the subdivision points (no constraint on those!). Then with

$$U = \max \{c_1, \dots, c_n, d_0, d_1, \dots, d_n\}$$

we get $U \geq t(x)$ for all x , and $t(x) \geq f(x)$, so $U \geq f(x)$. Running the same argument below f gives L . $\square$

Limits: the definition

Definition

Suppose $f: (a, b) \rightarrow \mathbb{R}$ and $c \in (a, b)$. We say $\lim_{x \rightarrow c} f(x) = L$ to mean: for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - c| < \delta \implies |f(x) - L| < \epsilon.$$

The $0 <$ (corrected from last time) says the definition is silent about $f(c)$ — indeed f need not be defined at c . That is what leaves **continuity** something to say.

Properties

If f and g are functions with $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, and $\lambda \in \mathbb{R}$, then

$$\blacklozenge \lim_{x \rightarrow c} (f + g)(x) = L + M$$

$$\blacklozenge \lim_{x \rightarrow c} fg = LM$$

$$\blacklozenge \lim_{x \rightarrow c} f/g = L/M, \text{ if } 1/g \text{ is defined (i.e. } g \neq 0)$$

$$\blacklozenge \lim_{x \rightarrow c} \lambda f = \lambda L$$

Where the product law comes from

Think of it as

$$f \sim L + (\text{bad}), \quad g \sim M + (\text{something}),$$

so that

$$fg \sim LM + \underbrace{(\text{bad})M + (\text{something})L + (\text{bad})(\text{something})}_{\text{error}}.$$

The job is to make the error small, which means estimating

$$|fg - LM|.$$

Useful identities

$$\diamond |A + B| \leq |A| + |B|$$

$$\diamond |AB| = |A| |B|$$

$$\begin{aligned} \diamond |A_1 B_1 - A_2 B_2| &= |A_1 B_1 - A_1 B_2 + A_1 B_2 - A_2 B_2| \\ &\leq |A_1| |B_1 - B_2| + |B_2| |A_1 - A_2| \end{aligned}$$

$$\diamond \text{ If } |A - B| < r \text{ then } |A| < |B| + r.$$

$$\text{Why? } |A| = |A - B + B| \leq |A - B| + |B| < |B| + r.$$

The product law

To show $\lim_{x \rightarrow c} f(x)g(x) = LM$, we need: for all $\epsilon > 0$ there is $\delta > 0$ such that $0 < |x - c| < \delta$ implies

$$|f(x)g(x) - LM| < \epsilon.$$

Pick $\epsilon > 0$. Since $\lim_{x \rightarrow c} f(x) = L$, we can choose δ_1 so that $0 < |x - c| < \delta_1$ gives

$$|f(x) - L| < \min \left\{ 1, \frac{\epsilon}{2|M|} \right\}.$$

Choosing the second δ

Choose δ_2 so that $0 < |x - c| < \delta_2$ gives

$$|g(x) - M| < \frac{\epsilon}{2(|L| + 1)},$$

and let $\delta = \min \{\delta_1, \delta_2\}$.

Note that $|f(x) - L| < 1$ already gives us

$$|f(x)| < |L| + 1,$$

which is the bound we need on the unruly factor.

Putting it together

For $0 < |x - c| < \delta$,

$$\begin{aligned} |f(x)g(x) - LM| &\leq |f(x)| |g(x) - M| + |M| |f(x) - L| \\ &< (|L| + 1) \cdot \frac{\epsilon}{2(|L| + 1)} + |M| \cdot \frac{\epsilon}{2|M|} \\ &= \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \quad \square \end{aligned}$$

(If $M = 0$ the second term is simply 0, and any bound will do for $|f(x) - L|$.)

Rephrasing

$$\lim_{x \rightarrow c} f(x)g(x) = \left(\lim_{x \rightarrow c} f(x) \right) \left(\lim_{x \rightarrow c} g(x) \right) \quad \text{if these exist}$$

The assumption of existence is necessary. For instance

$$\lim_{x \rightarrow 0} x g(x)$$

can perfectly well be 0 while $\lim_{x \rightarrow 0} g(x)$ does not exist at all — in which case splitting the product is not allowed, and the limit has to be found another way.

Using the limit properties

$$\lim_{x \rightarrow 3} x^2 = 3^2 \quad \text{because} \quad \lim_{x \rightarrow 3} x = 3.$$

In particular

If $f(x)$ is a polynomial or a rational function and f is defined at $x = c$, then

$$\lim_{x \rightarrow c} f(x) = f(c).$$

No ϵ , no δ — which is the whole point of having proved the laws.

A worked example

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{x^2 - 2x}{3 - x} &= \frac{\lim_{x \rightarrow 2} (x^2 - 2x)}{\lim_{x \rightarrow 2} (3 - x)} && \text{(if top and bottom limits exist)} \\ &= \frac{\lim_{x \rightarrow 2} x^2 - \lim_{x \rightarrow 2} 2x}{\lim_{x \rightarrow 2} 3 - \lim_{x \rightarrow 2} x} = \frac{(\lim x)^2 - 2(\lim x)}{\lim 3 - (\lim x)} \\ &= \frac{2^2 - 2(2)}{3 - 2} = 0.\end{aligned}$$

Continuity

Definition

If f is defined on an interval around c , we say f is **continuous at c** if

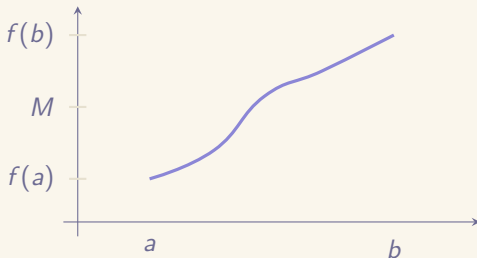
$$\lim_{x \rightarrow c} f(x) = f(c).$$

So the previous slide says exactly: **polynomials and rational functions are continuous wherever they are defined.**

The intermediate value theorem

Theorem

If $f: [a, b] \rightarrow \mathbb{R}$ is continuous and $f(a) < M < f(b)$, then there exists $c \in (a, b)$ such that $f(c) = M$.



Bolzano's theorem

Theorem

If $f: [a, b] \rightarrow \mathbb{R}$ is continuous with $f(a) < 0$ and $f(b) > 0$, then there exists $c \in (a, b)$ with $f(c) = 0$.

This is the special case $M = 0$ of the intermediate value theorem — and the general case follows from it by looking at $f(x) - M$.

A lemma

Lemma

If f is continuous at c and $f(c) < 0$, then $f(x) < 0$ for all $x \in (c - h, c + h)$, for some $h > 0$.

Proof. Let $\epsilon = |f(c)|$. Continuity at c gives a δ (this is our h) with

$$0 < |x - c| < \delta \implies |f(x) - f(c)| < \epsilon = |f(c)|.$$

Hence

$$f(x) \in (f(c) - |f(c)|, f(c) + |f(c)|),$$

and since $f(c) < 0$ we have $f(c) + |f(c)| = 0$, so $f(x) < 0$. $\square$

Proof of Bolzano's theorem

Suppose $f(a) < 0$ and $f(b) > 0$, and let

$$S = \{x \in [a, b] \mid f(x) < 0\}, \quad c = \sup(S).$$

If $f(c) < 0$: the lemma gives $f(c + h) < 0$ for some $h > 0$, so $c + h \in S$ with $c + h > c$ — contradicting $c = \sup(S)$. Hence $f(c) \geq 0$.

If $f(c) > 0$: the lemma (applied to $-f$) gives an $h > 0$ with $f(x) > 0$ for every $x \in (c - h, c + h)$. We derive a contradiction on the next slide.

Proof of Bolzano's theorem, concluded

So suppose $f(x) > 0$ for every $x \in (c - h, c + h)$.

Now $c - h < c = \sup(S)$. A number smaller than the *least* upper bound cannot itself be an upper bound — so $c - h$ is **not** an upper bound for S . By the definition of an upper bound, that means there is some $x \in S$ with

$$x > c - h.$$

And $x \leq c$, since c is an upper bound for S . So $x \in (c - h, c + h)$.

That x now satisfies both $f(x) > 0$ (it lies in the interval) and $f(x) < 0$ (it lies in S) — a contradiction.

Since $f(c)$ is neither negative nor positive, $f(c) = 0$. $\square$